

Rasmika Billa

☎ (551) 362-5882 | ✉ rb5719@nyu.edu | in rasmika-billa-b06b50199/ | 🌐 Rasmika-b

EDUCATION

NYU - Courant, New York

Sep 2024 – Present

Master of Science in Computer Science, AI Concentration, GPA: 3.7/4.0

Vellore Institute of Technology, India

Aug 2018 – Jul 2022

Bachelor of Technology in Computer Science, specialization in Data Analytics, GPA: 3.9/4.0

SKILLS

Languages: Python, R, SQL, MATLAB, Java, Pyspark, C/C++, JavaScript, HTML, CSS, TypeScript, React

AI/ML: PyTorch, Tensorflow, LangChain, NLP, Predictive Modelling, XGBoost, scikit-learn, NLTK, SpaCy

Developer Tools: Git, Docker, AWS, Google Cloud, Azure, Databricks, Linux, GitHub Actions, Kubernetes, CI/CD

Engineering Tools: ETL/ELT, DBT, Airflow, Power BI, Tableau, Snowflake, Databricks, Cursor

Frameworks/Databases: Flask, Streamlit, Next.js, Hadoop, Hive, Pinecone, BigQuery, MLOps, A/B Tests

EXPERIENCE

Research Assistant - NYU (*Predictive Analytics Lab*)

Nov 2024 – Present

- Quantified platelet activity from clinical data using **Quantile Regression**, reaching 78% prediction accuracy.
- Extracted return patterns from portfolio-level financial time-series data to support quantitative investment research.

AI Engineer Intern - Interplay (*Ascend FO Team*)

May 2025 – Aug 2025

- Deployed LLM-powered **Agentic AI** workflows to automate investment reporting across 14+ client portfolios, reducing turnaround time by 80%; translated portfolio manager requirements into production-grade data pipelines.
- Implemented **Differential Evolution**-based asset allocation optimization model to maximize Net-CAGR up to 8%.

Data Scientist - Royal Dutch Shell (*Asset Management Team*)

Aug 2022 – Aug 2024

- Engineered a **Retrieval Augmented Generation(RAG)** system using **LangChain**, custom API, and **SQL** in collaboration with safety and operations teams, achieving 86% model accuracy in report quality evaluation.
- Developed a chatbot using **GenAI(GPT API)** tailored to each terminal's safety data for efficient recommendation.
- Built data ingestion pipelines in Azure Data Factory for catalyst monitoring, contributing to a 4% revenue growth.
- Transformed unstructured data using **OCR**, reducing human error and saving 3,400+ work-hours at the plant site.

Data Analyst - Tendercuts (*Business Intelligence Team*)

Aug 2021 – Jul 2022

- Predicted weekly order volumes with **Decision Trees** for each store, leading to a 3% reduction in package usage.
- Conducted **RFM** analysis and built recommendation system that improved customer targeting effectiveness by 93%

RESEARCH/PATENT

AI For Factor Investing: A Comprehensive Survey (*Springer Nature, ICI AI 2025*)

Jan 2026

- Surveyed **ML**, **RL**, and **GenAI** methods for factor discovery, portfolio construction, trading, and risk control.
- Identified limitations of AI-based factor models and proposed strategies to address overfitting and interpretability.

Py - Digital Writing Pad (*Patent App no. 202141026740*)

Jun 2021

- Devised a motion-based tool with Transformer Architecture enabling users to capture ideas on-screen with any stylus.
- Achieved 72% reduction in costs by redesigning a pad utilizing **Deep Average Network** for hand motion detection.

Scheduling for Healthcare Centre for COVID-19 (*InHeNce - ID: 1570726833*)

June 2021

- Analyzed initial covid flow with **NetworkX** and produced a **Bi-LSTM** model with an MSE score of 7.2174e-04.
- Built a tool to optimize medical staff allocation using **Genetic** algorithm based on next week's predicted cases.

PROJECTS

Credit Guard Built and calibrated a credit risk model to predict 12-month loan default probability for a large Italian bank, achieving AUC **0.797** with **XGBoost** and improved robustness through influential observation analysis.

The Freebird Developed an app enabling users to buy or sell services, featuring a job recommendation system with high prediction accuracy using **RNN** (96.78% for skill level) and **Logistic Regression** (95.67% for job recommendations).

LifeTrack Performed survival analysis on patient data using the **Kaplan-Meier** estimator, log-rank test, and **Cox regression** model to predict event timing and mortality, and assess the impact of clinical variables on patient outcomes.